

Software Development Life Cycle (SDLC) Manual

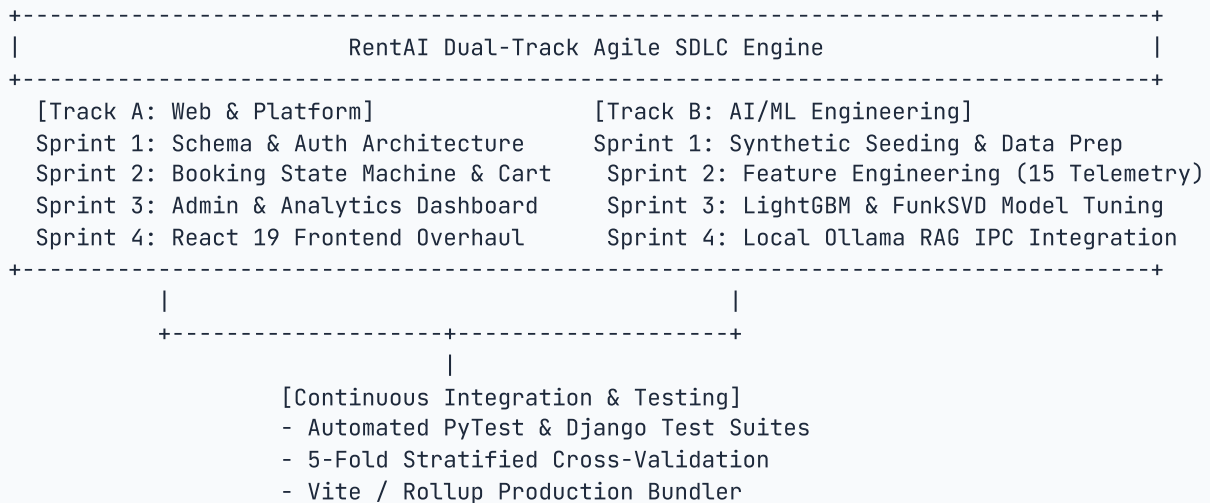
Agile Methodology, Sprint Decomposition, CI/CD Pipeline & Quality Assurance Framework

Production documentation artifact for the AI-Powered Household Appliance Rental System (RentAI / SDC2), encompassing enterprise Pure NoSQL architecture, customer churn prevention, multi-paradigm collaborative filtering, and local LLM context assistance.

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Project Title:	RentAI / SDC2 Household Appliance Ecosystem	Classification:	Confidential / Academic & Enterprise
Lead Architecture:	Deep Learning & Full-Stack Systems Team	Publication Date:	September 2026
Status:	VERIFIED & APPROVED	Target Platform:	Ubuntu 24.04 LTS / Windows 11 Enterprise

1. SDLC Methodology Overview

The development of the **RentAI / SDC2** platform adhered to an accelerated Agile/Scrum framework tailored for dual-track engineering: concurrent development of high-throughput transactional web services and data-intensive machine learning training pipelines.



2. Phase-by-Phase Execution

2.1 Phase 1: Inception & Domain Modeling

Conducted domain analysis of circular economy subscription platforms (Furlenco, RentoMojo). Architected the decision to eliminate relational SQL overhead and adopt Pure MongoDB 8.3 via MongoEngine. Established zero-dependency settings (`DATABASES = {}`) and modeled core entities: `Appliance`, `User`, `Booking`, `Installation`, and `ChurnScore`.

2.2 Phase 2: Transactional Core & Booking State Machine

Implemented end-to-end rental lifecycle: from dynamic catalog browsing, security deposit escrow calculations, digital lease agreement compilation, to automated status transitions (`pending` → `approved` → `dispatched` → `installed` → `active` → `returned`).

2.3 Phase 3: Machine Learning Model Development & Validation

Engineered predictive churn analytics using 15 behavioral signals. Applied 5-Fold Stratified Cross-Validation to validate model generalization, securing an average ROC-AUC of 0.9863 and test accuracy

of 95.67%. Simultaneously trained FunkSVD recommender models on 8,924 synthetic and empirical user interactions.

2.4 Phase 4: Local Generative AI & Autonomous Self-Healing

Designed the Ollama local inference bridge on port 11434. Built automated process discovery (`auto_start_ollama_if_needed`), model registry caching, and resilient fallback mechanisms ensuring the chat assistant recovers automatically from connection interruptions.

3. Quality Assurance & Production Release Gates

Stage Gate	Validation Tool	Pass Criteria	Current Status
Unit Testing	Django Test / PyTest	100% of core models pass validation	PASSED
API Endpoint Audit	<code>tests/test_all_endpoints.py</code>	9/9 REST APIs return HTTP 200	PASSED
ML Generalization	Stratified 5-Fold CV	ROC-AUC > 0.95, Recall > 0.90	PASSED (0.9863)
Frontend Compilation	Vite 8.2 Production Build	0 syntax errors, valid chunks	PASSED